Code No: 157EQ **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
FUNDAMENTALS OF BIOMEDICAL APPLICATIONS
(Common to ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Discuss what are the problems encountered in measurement of biological parameters.
b) Derive Nernst equation for membrane resting potential. [8+7]

2.a) Draw the block diagram of biomedical instrumentation system and explain the functions of each block.
b) Discuss the dynamic characteristics of Medical Instruments. [8+7]

3.a) What is a Recorder? With a neat diagram explain the principle of working of an Inkjet Recorder.
b) Give the origin, amplitude and duration of different waves in ECG signal. [8+7]

4.a) Clearly explain the standard 12-lead configuration in ECG Wave form recording.
b) Discuss the effects of artifacts on ECG recording. [8+7]

5.a) Sketch the block diagram of automated electro sphygmomanometer for blood pressure measurement and explain its operation.
b) Explain the working principle of phonocardiograph with sound waves. [8+7]

6.a) Describe the method of measuring Blood Pressure by cardiac catheterisation.
b) Discuss with necessary diagram Doppler Imaging system for blood flow measurement.[7+8]

7.a) State the need for defibrillator. Describe the schematic of implantable defibrillator.
b) What is meant by Electrotherapy? Also discuss its applications. [8+7]

8.a) Explain with neat diagram the working of EMG.
b) Draw the block diagram of a ventilator along with its accessories and explain its function. [7+8]

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